

The presentation of mathematics in schools should be psychological and not systematic. The teacher should be a diplomat. He must take account of the psychic processes in the boy in order to grip his interest, and he will succeed only if he presents things in a form intuitively comprehensible. A more abstract presentation is only possible in the upper classes.

Felix Klein, quoted in MacHale (1993)

In the previous chapter, we derived the 32 point group symmetries that are compatible with the translational symmetry of the 14 Bravais lattices. In the present chapter we ask the next logical question: *what happens when we place a molecule (or a motif) with a certain point group symmetry $\mathcal{G}$ on each lattice node of a certain Bravais lattice $\mathcal{T}$?* We will show that this leads to the development of the 230 3-D crystallographic space groups; in 2-D, there are 17 plane groups. Furthermore, when time-reversal symmetry is included, the total numbers of plane and space groups increase dramatically to 80 and 1651, respectively. We conclude this chapter with a discussion of the use of space groups based on the *International Tables for Crystallography*.

10.1

Combining translations with point group symmetry

To answer the question above fully, we need to take every point group that belongs to a given crystal system and combine it with the translational symmetries of each of the Bravais lattices belonging to the same crystal system. For instance, for the cubic point group $\mathbf{m}\bar{3}\mathbf{m}$ (O_h), we would need to combine it with the three cubic Bravais lattices cP , cI , and cF . For each of these combinations, we need to ask the question: is this a new symmetry group? Furthermore, for each combination we would need to replace each mirror plane by all possible compatible glide planes, and each rotation axis by all possible screw axes and again ask the question: is this a new symmetry group?

The reader might wonder where all these new symmetry elements come from? Consider a simple example. If we have two parallel mirror planes, $\mathbf{m}_1$ and $\mathbf{m}_2$, separated by a distance $t/2$, as shown in Fig. 10.1, then a consecutive mirror operation in both mirrors is easily seen to be equivalent to a translation over a distance t . Object 1 is mirrored into object 2 by the first mirror, and then again mirrored into object 3 by the second mirror. The distance between objects 1 and 3 is equal to twice the distance between the mirror planes. Conversely, since the combination of any two symmetry operations must again be

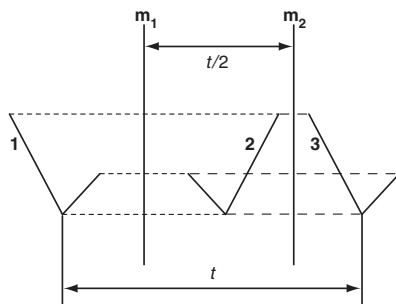


Figure 10.1 Illustration of the equivalence of two parallel mirror planes at a distance $t/2$ from each other to a lattice translation by a distance t .

a symmetry operation, we also find that a mirror plane combined with a lattice translation perpendicular to that mirror plane *must* give rise to a new mirror plane at a distance half that of the translation length. This simple example illustrates that, as we combine symmetry operations from the point groups with lattice translations, we will create many additional symmetry operations. Glide planes and screw axes are amongst these newly generated symmetry elements.

Figuring out which of these combinations gives rise to a new symmetry group is clearly a formidable task. This task was first completed by the end of the nineteenth century by Federov in Russia (Federov, 1891), Barlow in England (Barlow, 1894), and Schoenflies in Germany (Schoenflies, 1891). They independently concluded that there are “only” 230 distinct symmetries. This is a very important result! It essentially tells us that *every* crystal in nature, and *every* crystalline material we can fabricate, *must* have one of those 230 symmetries. Note that this result was obtained by pure group theoretical reasoning, and that very few, if any, experimental observations were used in its derivation. The 230 resulting symmetry groups are known as the 3-D *space groups*.

Instead of systematically enumerating all 230 space groups, a somewhat lengthy task, we will, in this chapter, present a few examples of space group symmetry in 2-D and 3-D, and explain the international notation for the space groups. Then we will discuss the two types of space group (symmorphic and non-symmorphic), we will illustrate how one can obtain the magnetic space groups, and we will conclude with a description of the space group entries in the *International Tables for Crystallography*, volume A.

10.2

Plane groups

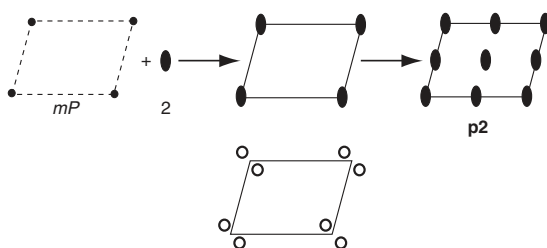
10.2.1

A simple example

Since there are only ten 2-D point groups, and five different 2-D Bravais lattices, it should not be too difficult to consider all possible combinations. A *plane group* is the infinite

Table 10.1 2-D Bravais lattices, point groups, and the number of plane groups for each Bravais lattice type.

2-D Bravais lattice	Point groups	# plane groups
Primitive oblique	1 (C_1), 2 (C_2)	2
Primitive rectangular	m (C_s), mm2 (C_{2v})	5
Centered rectangular	m (C_s), mm2 (C_{2v})	2
Primitive square	4 (C_4), 4mm (C_{4v})	3
Primitive hexagonal	3 (C_3), 3m (C_{3v}), 6 (C_6), 6mm (C_{6v})	5

**Figure 10.2** Construction of the **p2** plane group, one of the 17 2-D plane groups from the primitive oblique Bravais lattice and the point group **2** (C_2).

group obtained by combining point group symmetries (and glides) with the translational symmetries of a 2-D lattice. The plane groups are obtained as the union of the translational symmetries of the 2-D lattice with the point group symmetry about the lattice point when decorated with an atomic or molecular basis.

The assignment of the ten point groups to their respective crystal system results in ten plane groups determined by their combination with the primitive Bravais lattice in each system. Two additional plane groups are obtained by combining **m** (C_s) and **mm2** (C_{2v}) with the centered rectangular lattice. The last five plane groups are determined by adding glide operations where compatible with the preceding 12 plane groups. This yields 17 plane groups with their distribution among the five 2-D Bravais lattices, as summarized in Table 10.1.

As an example of a procedure to determine the symmetry operations of a plane group, we consider Fig. 10.2. This figure illustrates the decoration of an oblique lattice, *mP*, with objects having the **2** (C_2) 2-D point group symmetry. First we draw a primitive oblique unit cell. Then we consider the **2** (C_2) point group, represented graphically by the $\bullet$ symbol. This point group has two operations, the identity and a two-fold proper rotation. At every lattice point of the oblique cell, we place a copy of the point group symbol. Then we look for all implied symmetry elements. If we were to take an arbitrary

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object at a point (x, y) near a lattice point $(0, 0)$, the action of the two-fold rotation operator is to replicate it at the position $(-x, -y)$. A translation along the a -axis (horizontal direction) would then replicate it at the position $(1 - x, -y)$. An equivalent operation that takes an object from the position (x, y) to the position $(1 - x, -y)$ is a two-fold rotation about the point $(1/2, 0)$. Therefore, we conclude that this plane group has two-fold rotation operators at all cell edge centers. Hence, we draw the $\bullet$ symbol at each edge center. A similar analysis allows us to conclude that the cell center also has two-fold symmetry, so we draw the $\bullet$ symbol there as well. This exhausts all of the symmetries of this plane group.

The result of this derivation is the plane group **p2**. Note the nomenclature used for the plane groups: first we use the Bravais lattice centering symbol (**p** for “primitive” in this case), followed by the point group symbol (**2**). Note that the oblique lattice is the only primitive lattice that we can combine with the **2** (C_2) point group. Therefore, it is not necessary to state the crystal system explicitly in the plane group symbol **p2**.

The prior procedure can also be depicted in terms of how a general point is “copied” into equivalent points in the lattice. This is shown in the bottom sketch of Fig. 10.2. Here an object, represented by an open circle, is placed near a lattice point. The action of the two-fold operator replicates this object at a position rotated by an angle π about this point. The pair of objects is then replicated by adding translations to repeat it within the cell and adjacent cells. After this is complete, it is apparent that the resulting collection of objects possesses two-fold symmetries about the cell center and edge centers, in addition to the two-fold symmetry about the vertices of the cell. Note that the orbit of a general point (i.e., the set of all equivalent points) in a plane group is an infinite set of points, because the lattice translations continue from $-\infty$ to $+\infty$ in both directions. Note also that there are only two equivalent points per unit cell.

The *site symmetry* of a general point is by definition equal to **1** (C_1). In the plane group **p2**, there are four special positions, namely the locations of the two-fold axes, $(0, 0)$, $(0, 1/2)$, $(1/2, 0)$, and $(1/2, 1/2)$. The site symmetry at each of these locations is, obviously, **2** (C_2). The Wyckoff positions are then *1a*, *1b*, *1c*, and *1d*, respectively for the four locations of the two-fold axes, and *2e* for the general position (x, y) . The *International Tables of Crystallography* (Hahn, 1996) lists all of the general and special points for all 17 plane groups (Chapter 6).

10.2.2

A more complex example

Let us consider another example of the construction of a plane group. Figure 10.3 illustrates the decoration of a square lattice with objects having the **4mm** (C_{4v}) point group symmetry. This results in the **p4mm** plane group, as follows. First, we draw a primitive square unit cell, tP . Then we consider the **4mm** (C_{4v}) point group. To the $\blacklozenge$ symbol for a four-fold rotation axis we add the four mirror planes to represent the 2-D point group. This point group has eight operations: the identity, the three four-fold proper rotation operations, and four mirror lines. If we take an arbitrary object at a point (x, y) near the lattice point $(0, 0)$, the action of the four-fold rotation operator is to replicate it at the positions $(-y, x)$,

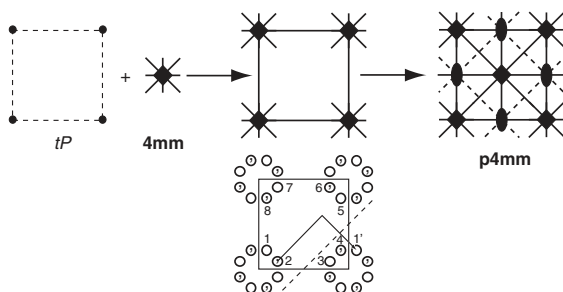


Figure 10.3 Construction of **p4mm** plane group, one of the 17 2-D plane groups from the tP Bravais lattice and the point group **4mm** (C_{4v}).

$(-x, -y)$, and $(y, -x)$, respectively. The mirror lines give rise to the positions $(-x, y)$, $(x, -y)$, (y, x) , and $(-y, -x)$.

We replicate the point group symbol at every lattice point of the square cell. We align the mirror lines along the cell edges and the cell diagonals. Then we look for all implied symmetry elements. It is clear that the square lattice must have a four-fold axis at the center of each cell, so we draw the $\blacklozenge$ symbol at the cell center. The mirrors along the cell diagonal pass through the center as well. The four-fold axis at the center combines with the two diagonal mirrors to produce two new mirrors, which pass through the center of the cell edges. This indicates that the point at the center of the cell also has **4mm** (C_{4v}) point group symmetry. Since each edge has a pair of intersecting mirrors, there must be a two-fold axis at each edge center. The presence of parallel diagonal mirror lines through the origin and the cell center implies the presence of another mirror line centered halfway between the original two mirror lines. This new diagonal mirror line turns out to be a glide line; this is easy to show when we consider the bottom sketch of Fig. 10.3. Here an object, represented by an open circle, is placed near a lattice point. The action of the four-fold operator replicates this object at positions rotated by $\pi/2$, π , and $3\pi/2$ about this point. The mirror lines replicate each of the four right-handed objects into left-handed objects, represented by an open circle with a comma in the center. The eight objects are then replicated by adding translations to repeat them within the cell and adjacent cells. We see that there are eight equivalent points inside the unit cell. There must be precisely one symmetry element connecting each pair of two equivalent points. For instance, point 1 is converted into point 3 by the four-fold rotation at the center of the cell, and points 5 and 7 through the repeated application of the four-fold rotation. Point 1 can be mapped onto point 1' by a regular lattice translation along the horizontal direction. We can also reach point 1' starting from point 2, by means of a diagonal glide line going through the points with coordinates $(1/2, 0)$ and $(1, 1/2)$ (dashed line). The hooked arrow indicates the translate-mirror motion of point 2. It is clear from this simple illustration that in an infinite space group, there is precisely one symmetry operation for each pair of points, no matter how many unit cells apart these points are. This completes the derivation of the **p4mm** plane group. Note that once again the crystal system, t , is not mentioned in the plane group symbol, because the point group **4mm** (C_{4v}) can only be combined with the square lattice.

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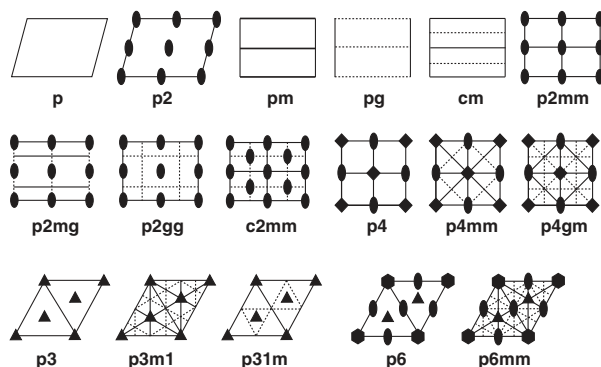


Figure 10.4 The 17 2-D plane groups: **p1**, **p2**, **pm**, **pg**, **cm**, **p2mm**, **p2mg**, **p2gg**, **c2mm**, **p4**, **p4mm**, **p4gm** **p3**, **p3m1**, **p31m**, **p6**, and **p6mm**.

10.2.3

The 17 plane groups

The same procedure can be applied to derive the other 15 plane groups. Without going into all the details, here is how the general derivation works. We start by considering the most general (lowest symmetry) of the 2-D Bravais lattices, the oblique lattice, which is simply a parallelogram with angles between the bases that are not $\pi/2$ or $2\pi/3$. This lattice is compatible only with the **1** (C_1) and **2** (C_2) point groups. Combination of **1** (C_1) with the oblique lattice gives rise to the primitive plane group labeled **p1**. Combination of **2** (C_2) with the oblique lattice gives rise to the plane group **p2** (also primitive) which was derived. Figure 10.4 shows the crystallographic representation of all 17 plane groups. As in Figs. 10.2 and 10.3, this consists of (a) decoration of the lattice points with the appropriate point group operations; and (b) inclusion of new symmetry elements implied by the first. For additional illustrations and lists of general and special point positions the reader is referred to Chapter 6 in the *International Tables for Crystallography* (Hahn, 1996).

Combination of a mirror plane with the rectangular lattice gives rise to the primitive plane group, **pm**. Replacing the mirror line by an axial glide line, we find a second plane group in the rectangular Bravais lattice: **pg**. Combination of a mirror with the centered rectangular lattice also gives rise to a primitive plane group; however, it is by convention represented as an equivalent centered rectangular plane group with symbol **cm**. It is also possible to combine two orthogonal mirror planes with the rectangular Bravais lattice giving rise to the **p2mm** plane group; additional two-fold axes are required at the intersections between the orthogonal mirrors. Single and double axial glides give rise to the **p2mg** and **p2gg** plane groups, respectively. The centered rectangular lattice produces axial glide planes and results in the **c2mm** plane group.

The plane group **p4** is obtained by adding the **4** (C_4) point group to the square Bravais lattice. The **p4mm** plane group, derived above, has two orthogonal sets of diagonal glide lines that are implied and therefore not included in the plane group symbol. Addition of new axial and diagonal glide lines to the group **p4** gives rise to the plane group **p4gm**.

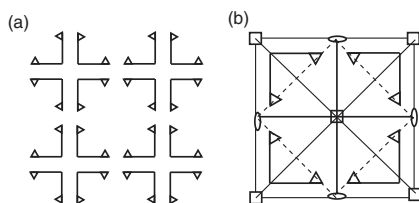


Figure 10.5 (a) Kockel diagram of a symmetric 2-D structure and (b) its unit mesh and symmetry operations.

Three- and six-fold rotational axes are only compatible with the hexagonal Bravais lattice. The primitive unit cell is represented as a parallelogram (rhombus). In plane groups with three-fold or six-fold rotational symmetry, the rotation operation is replicated at all lattice sites. In both cases, three-fold axes are present also at positions $(1/3, 2/3)$ and $(2/3, 1/3)$ along the cell diagonal.

Combination of the hexagonal lattice and a three-fold axis gives rise to the primitive plane group **p3**. There are two ways of adding mirror planes to the plane group **p3**. The three-fold axis replicates the mirror planes along the edges of an underlying triangular lattice. The plane group **p3m1** has an additional mirror line along the cell diagonal, mirror lines passing through the cell vertices and the midpoint of opposite edges as well as implied glide lines. The **p31m** plane group does not have the additional mirror lines but does have a triangular network of glide lines around the three-fold axes located on the cell diagonal. Combination of the triangular lattice and a six-fold axis gives rise to the primitive group **p6**. Addition of a mirror plane to **p** gives rise to the plane group **p6mm**. In both cases, two-fold axes on the cell edges are implied.

As a final exercise, we consider the 2-D structure illustrated in Fig. 10.5(a). This is an example of a *Kockel diagram* (Borchardt-Ott, 1995), which can be used to illustrate the symmetry of a particular plane group using a structural motif. Such diagrams can provide interesting exercises to test the comprehension of the subject. In such an exercise, one is typically asked to (1) determine the unit cell (unit mesh); (2) show all symmetry elements in the cell; and (3) determine the 2-D plane group for the structure. The example in Figure 10.5 is a square mesh that has four-fold and two-fold rotation axes, vertical and diagonal mirror lines, and diagonal glide lines. It is, therefore, an example of the **p4mm** plane group derived in Section 10.2.2.

10.3

Space groups

10.3.1

A simple example

Consider the 3-D orthorhombic point group **mm2** (C_{2v}). It has four elements: the identity, a two-fold axis parallel to the z -direction, and two mirror planes m_1 and m_2 at right angles to each other. Since orthorhombic point groups can only be compatible with the orthorhombic

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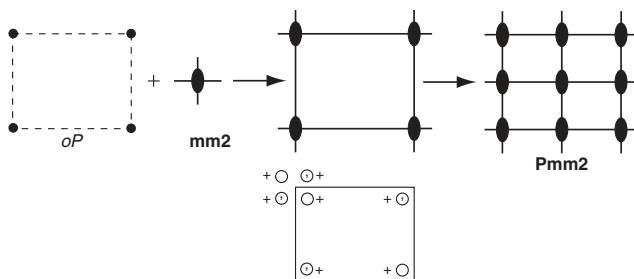


Figure 10.6 Construction of the space group $\text{Pmm}2$ (C_{2v}^1) from the Bravais lattice oP and the point group $\text{mm}2$ (C_{2v}).

Bravais lattices (of which there are four), we need to investigate the combination of these symmetry elements with the various orthorhombic centering operators. Figure 10.6 shows how we can construct a space group based on the point group $\text{mm}2$ (C_{2v}) and the Bravais lattice oP . At every lattice point of oP , indicated by filled circles on the drawing, we put a copy of the point group drawing. We notice that the mirror planes line up with the unit cell boundaries, so we draw solid lines to indicate that they are indeed mirror planes. We also know that two parallel mirror planes at a distance t give rise to an additional mirror plane at the distance $t/2$. Hence we draw two new mirror planes halfway through the unit cell. Finally, we know that the intersection of two mirror planes is equivalent to a two-fold axis along the intersection line, so we draw the $\bullet$ symbol at each intersection. This completes the drawing. This configuration of symmetry elements is, of course, repeated in every unit cell. The infinite set of symmetry elements thus obtained forms the *space group* $\text{Pmm}2$ (C_{2v}^1). The space group symbol is formed by combining the centering information of the Bravais lattice with the point group symbol (in international or Hermann–Mauguin notation). It is not necessary to include the o for orthorhombic, since the point group symbol already indicates that the group must have orthorhombic symmetry. Note that the centering symbol is written as an upper-case letter, as opposed to the centering symbols for the 2-D plane groups, which are written as lower-case letters. The Schoenflies notation for the space groups will also be shown between parentheses; this symbol is based on the point group Schoenflies symbol.

Finally, we can indicate how a general point is “copied” into the equivalent points. This is shown in the bottom sketch of Fig. 10.6. The open circle with the $+$ sign next to it indicates a general point *above* the plane of the drawing, a minus sign would be below. The symbol $\frac{1}{2}+$ indicates that the point is half a unit cell above the original point, etc. The circles with a comma inside indicate that the point is related to the original point by an improper symmetry operation. In other words, those points are left-handed versions of the original right-handed point. This type of drawing is for space groups what the stereographic representations are for point groups.

The space group $\text{Pmm}2$ (C_{2v}^1) has four equivalent points inside the unit cell; these points have coordinates:

$$(x, y, z), \quad (\bar{x}, \bar{y}, z), \quad (x, \bar{y}, z), \quad (\bar{x}, y, z).$$

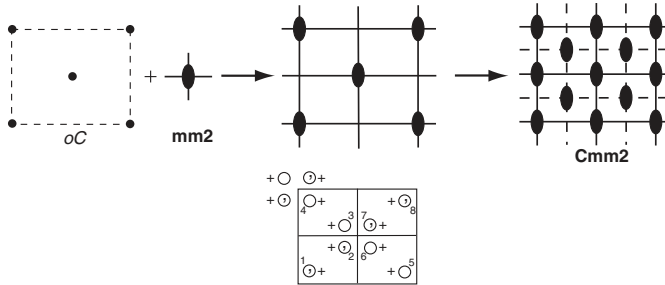


Figure 10.7 Construction of the space group **Cmm2** (C_{2v}^{11}) from the Bravais lattice oC and the point group **mm2** (C_{2v}).

The positions with the highest site symmetry are found along the two-fold rotation axes, located at $(0, 0, z)$, $(0, 1/2, z)$, $(1/2, 0, z)$, and $(1/2, 1/2, z)$. These four locations have site symmetry **mm2**, and are denoted by the Wyckoff positions $1a$, $1b$, $1c$, and $1d$, respectively. In addition, there are four mirror planes in this space group, leading to the following special positions: $(x, 0, z)$, $(x, 1/2, z)$, $(0, y, z)$, and $(1/2, y, z)$. The site symmetry for the first two of these is **m**, and for the other two it is **m**. The Wyckoff positions are labeled $2e$, $2f$, $2g$, and $2h$, respectively. The general position, (x, y, z) , then gets the notation $4i$.

10.3.2

A second simple example

Let us repeat this exercise for the orthorhombic oC lattice, using the same point group symmetry. Figure 10.7 shows the steps in the construction of space group **Cmm2** (C_{2v}^{11}). The centering vector of the oC Bravais lattice copies every point with coordinates (x, y, z) onto an equivalent point $(x + 1/2, y + 1/2, z)$. Looking at the equivalent sites inside the unit cell (bottom sketch in Fig. 10.7), we find that there are eight equivalent positions. If we take any pair of points, there must be a symmetry operator that converts one into the other. For instance, point 1 is converted into point 2 by means of the two-fold rotation axis located at position $(1/4, 1/4, 0)$; 2 goes to 7 by means of the two-fold rotation axis at the center of the cell, as do 4 to 5, 1 to 8, and 3 to 6. Which symmetry operator brings 1 onto 3? To answer this question we start from the presence of two parallel horizontal mirror planes (second sketch) going through the points $(0, 0, 0)$ and $(1/2, 0, 0)$. Owing to the translational symmetry, there must be a third parallel mirror plane halfway in between them, going through the point $(1/4, 0, 0)$. From the positions of the equivalent points, we see that this mirror plane is not just a simple mirror plane, but instead a glide plane with a translation vector $(0, 1/2, 0)$. This glide plane takes point 1, translates it to the position $(x, y + 1/2, z)$, and then mirrors it into position 3, with coordinates $(1/2 - x, y + 1/2, z)$.

The full set of eight equivalent points has the following coordinates:

$$(0, 0, 0) + \left(\frac{1}{2}, \frac{1}{2}, 0 \right) +$$

$$(x, y, z) \quad (\bar{x}, \bar{y}, z) \quad (x, \bar{y}, z) \quad (\bar{x}, y, z).$$

10.3 Space groups

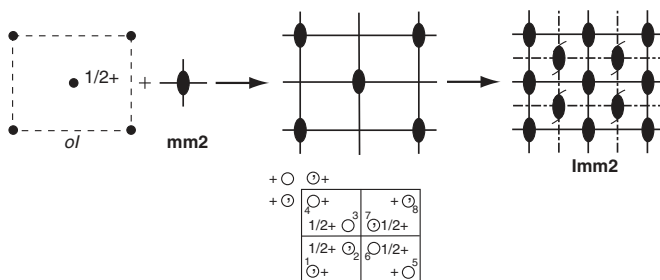


Figure 10.8 Construction of the space group **Imm2** (C_{2v}^{20}) from the Bravais lattice *ol* and the point group **mm2** (C_{2v}).

The first line contains the zero-vector and the centering vector **C**. To find all the equivalent point coordinates we must add these vectors (hence the + sign) to all four coordinates on the second row. Those coordinates are found by matrix multiplication of the symmetry matrices of the point group **mm2** (C_{2v}) with the general position vector (x, y, z) . The *multiplicity* of a general point is equal to the number of equivalent positions for a general point inside a single unit cell. For this space group, **Cmm2** (C_{2v}^{11}), the multiplicity is eight, whereas it is four for **Pmm2** (C_{2v}^1). Note that the multiplicity for a space group is similar to the *order* for a point group; since all space groups are of infinite order, there is no point in using the order of the space group to say something about the number of equivalent points inside a single unit cell.

10.3.3

A more complex example

As a third example we will combine the Bravais lattice *ol* with the point group **mm2** (C_{2v}). The construction is shown in Fig. 10.8 and results in the space group **Imm2** (C_{2v}^{20}). We must remember that the point group symbol is copied at every lattice site, but now the central lattice site is at a position $1/2$ unit cell *above* the plane of the drawing. The two-fold axis, which for the space group **Cmm2** (C_{2v}^{11}) is generated at the position $(1/4, 1/4, 0)$ now becomes a two-fold screw axis, indicated by the symbol $\uparrow$. This is easy to understand when we consider how to go from point 1 to point 2: we rotate 180° around the point $(1/4, 1/4, 0)$, and then we must translate by $(0, 0, 1/2)$ along the axis to obtain point 2. This combined operation is a 2_1 screw axis. A similar analysis for the points 1 and 3 reveals that the glide plane must be a diagonal glide rather than the axial glide that we found for space group **Cmm2** (C_{2v}^{11}). The dash-dotted lines represent the four *diagonal* glide planes in the unit cell. The multiplicity of a general point is again equal to eight, but the coordinates are now given by:

$$(0, 0, 0) + \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) +$$

$$(x, y, z) \quad (\bar{x}, \bar{y}, z) \quad (x, \bar{y}, z) \quad (\bar{x}, y, z).$$

The special positions for this space group are easy to determine: the highest symmetry sites are located at the intersections of two mirrors or glides and a two-fold rotation or screw axis. Those are the positions $(0, 0, z)$ and $(0, 1/2, z)$, which receive the Wyckoff symbols $2a$ and $2b$, respectively. Then there are the mirror planes normal to the $\mathbf{b}$ and $\mathbf{a}$ basis vectors, which correspond to positions of the types $4c$: $(x, 0, z)$ and $4d$: $(0, y, z)$, respectively. The general position (x, y, z) receives the Wyckoff symbol $8e$.

We have found that the space group **Cmm2** (C_{2v}^{11}) contains axial glide planes amongst its elements, and that **Imm2** (C_{2v}^{20}) has two-fold screw axes and diagonal glide planes. Those symmetry elements are the result of the combinations of other, “regular,” symmetry elements, and therefore we do not need to include the symbols for the axial glides or the screw-rotation symbol explicitly in the space group symbol. If a space group symbol does not contain any screw-rotation symbols or glide plane symbols, then that space group is known as a *symmorphic* space group. If the symbol does contain either screw-rotation symbols or glide plane symbols, or any combination of both, then that space group is known as a *non-symmorphic* space group. Examples of the latter type include **Cmc2₁** (C_{2v}^{12}), **P4₂nm** (C_{4v}^4), **I4/mcm** (D_{4h}^{18}), and **Fd3m** (O_h^7). In the following section, we will discuss the symmorphic space groups in more detail. In [Section 10.3.5](#) we will describe the non-symmorphic groups.

10.3.4

The symmorphic space groups

As mentioned, a symmorphic space group is a space group with a symbol that does not contain any screw-rotations or glide planes. One can enumerate all the symmorphic space groups by taking the combinations of all Bravais lattices with all point groups of the same crystal class, just as we did in the previous section. In other words, for the lattice aP we have two triclinic point groups, $\mathbf{1}$ (C_1) and $\bar{\mathbf{1}}$ (C_i); for mP there are three monoclinic point groups, for mC there are also three monoclinic point groups, and so on. This results in the formation of 61 space groups, listed in [Table 10.2](#).¹ Note that each space group has been assigned a sequential number, from 1 to 230. The international notation is shown along with the Schoenflies notation; it is clear that the Schoenflies notation is simply a sequential numbering added to the original point group Schoenflies symbol. These symbols are not as instructive as the Hermann–Mauguin symbols.

This does not conclude the enumeration of the symmorphic space groups, however. There are 12 additional symmorphic space groups, indicated in [Table 10.2](#) by an underlined space group number. The additional space groups are formed in a number of different ways:

- We have assumed that there is only one possible *relative orientation* between the point group symmetry elements and the Bravais lattice directions. For several point groups there is more than one way to copy the point group onto the lattice. A typical example would be the space groups **P4₂m** (D_{2d}^1) and **P4m2** (D_{2d}^5). For space group **P4₂m** (D_{2d}^1), the mirror planes are positioned along the diagonals of the tetragonal unit cell, whereas

¹ It is easy to verify that there are 61 groups resulting from the combination of Bravais lattices and point groups: 1×2 (triclinic) + 2×3 (monoclinic) + 4×3 (orthorhombic) + 2×7 (tetragonal) + 1×5 (trigonal) + 1×7 (hexagonal) + 3×5 (cubic) = 61.

Table 10.2 The 73 symmorphic space groups, with sequential number, international and Schoenflies symbols, and corresponding point group.

Space group #	Point group	Symbol	Space group #	Point group	Symbol
1	1 (C_1)	P1 (C_1^1)	143	3 (C_3)	P3 (C_3^1)
2	$\bar{1}$ (C_i)	P$\bar{1}$ (C_i^1)	<u>146</u>		R3 (C_3^4)
3	2 (C_2)	P2 (C_2^1)	147	$\bar{3}$ (C_{3i})	P$\bar{3}$ (C_{3i}^1)
5		C2 (C_2^3)	<u>148</u>		R$\bar{3}$ (C_{3i}^2)
6	m (C_s)	Pm (C_s^1)	149	32 (D_3)	P312 (D_3^1)
8		Cm (C_s^3)	<u>150</u>		P321 (D_3^2)
10	2/m (C_{2h})	P2/m (C_{2h}^1)	<u>155</u>		R32 (D_3^3)
12		C2/m (C_{2h}^3)	156	3m (C_{3v})	P3m1 (C_{3v}^1)
16	222 (D_2)	P222 (D_2^1)	<u>157</u>		P31m (C_{3v}^2)
21		C222 (D_2^6)	<u>160</u>		R3m (C_{3v}^5)
22		F222 (D_2^7)	162	$\bar{3}m$ (D_{3d})	P$\bar{3}1m$ (D_{3d}^1)
23		I222 (D_2^8)	<u>164</u>		P$\bar{3}m1$ (D_{3d}^3)
25	mm2 (C_{2v})	Pmm2 (C_{2v}^1)	<u>166</u>		R$\bar{3}m$ (D_{3d}^5)
35		Cmm2 (C_{2v}^{11})	168	6 (C_6)	P6 (C_6^1)
<u>38</u>		Amm2 (C_{2v}^{14})	174	$\bar{6}$ (C_{3h})	P$\bar{6}$ (C_{3h}^1)
42		Fmm2 (C_{2v}^{18})	175	6/m (C_{6h})	P6/m (C_{6h}^1)
44		Imm2 (C_{2v}^{20})	177	622 (D_6)	P622 (D_6^1)
47	mmm (D_{2h})	Pmmm (D_{2h}^1)	183	6mm (C_{6v})	P6mm (C_{6v}^1)
65		Cmmm (D_{2h}^{19})	187	$\bar{6}m2$ (D_{3h})	P$\bar{6}m2$ (D_{3h}^1)
69		Fmmm (D_{2h}^{23})	<u>189</u>		P$\bar{6}2m$ (D_{3h}^3)
71		Immm (D_{2h}^{25})	191	6/mmm (D_{6h})	P6/mmm (D_{6h}^1)
75	4 (C_4)	P4 (C_4^1)	195	23 (T)	P23 (T^1)
79		I4 (C_4^5)	196		F23 (T^2)
81	$\bar{4}$ (S_4)	P$\bar{4}$ (S_4^1)	197		I23 (T^3)
82		I$\bar{4}$ (S_4^2)	200	m$\bar{3}$ (T_h)	Pm$\bar{3}$ (T_h^1)
83	4/m (C_{4h})	P4/m (C_{4h}^1)	202		Fm$\bar{3}$ (T_h^3)
87		I4/m (C_{4h}^5)	204		Im$\bar{3}$ (T_h^5)
89	422 (D_4)	P422 (D_4^1)	207	432 (O)	P432 (O^1)
97		I422 (D_4^9)	209		F432 (O^3)
99	4mm (C_{4v})	P4mm (C_{4v}^1)	211		I432 (O^5)
107		I4mm (C_{4v}^9)	215	$\bar{4}3m$ (T_d)	P$\bar{4}3m$ (T_d^1)
111	$\bar{4}2m$ (D_{2d})	P$\bar{4}2m$ (D_{2d}^1)	216		F$\bar{4}3m$ (T_d^2)
<u>115</u>	4m2 (D_{2d})	P4m2 (D_{2d}^5)	217		I$\bar{4}3m$ (T_d^3)
<u>119</u>	4m2 (D_{2d})	I4m2 (D_{2d}^9)	221	m$\bar{3}m$ (O_h)	Pm$\bar{3}m$ (O_h^1)
121	$\bar{4}2m$ (D_{2d})	I$\bar{4}2m$ (D_{2d}^{11})	225		Fm$\bar{3}m$ (O_h^5)
123	4/mmm (D_{4h})	P4/mmm (D_{4h}^1)	229		Im$\bar{3}m$ (O_h^6)
139		I4/mmm (D_{4h}^{17})			

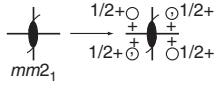


Figure 10.9 The symmetry combination $mm2_1$ gives rise to eight equivalent points, which correspond to twice the arrangement of points for the space group $\mathbf{Pmm2}$ (C_{2v}^1).

for group $\mathbf{P}\bar{4}m2$ (D_{2d}^5) they are parallel to the faces of the unit cell. This is indicated by reversing the order of the two-fold axis and the mirror plane in the Hermann–Mauguin symbol. A similar situation occurs for space groups $\mathbf{I}\bar{4}m2$ (D_{2d}^9) and $\mathbf{I}\bar{4}2m$ (D_{2d}^{11}), and for $\mathbf{P6m2}$ (D_{3h}^1) and $\mathbf{P62m}$ (D_{3h}^3).

- For the orthorhombic point group $\mathbf{mm2}$ (C_{2v}) we can position the two-fold axis perpendicular to the C -centered plane, which results in $\mathbf{Cmm2}$ (C_{2v}^{11}), or we can position the two-fold axis parallel to the centered plane, in which case one uses the A -centered unit cell. The latter case gives rise to the symmorphic space group $\mathbf{Amm2}$ (C_{2v}^{14}).
- The trigonal point groups can be combined with the R Bravais lattice, which gives rise to the space groups $\mathbf{R3}$ (C_3^4), $\mathbf{R}\bar{3}$ (C_{3i}^2), $\mathbf{R32}$ (D_3^7), $\mathbf{R3m}$ (C_{3v}^5), and $\mathbf{R}\bar{3}m$ (D_{3d}^5). The trigonal point groups can also be combined with a hexagonal primitive Bravais lattice, which gives rise to space groups $\mathbf{P312}$ (D_3^1), $\mathbf{P3m1}$ (C_{3v}^1), and $\mathbf{P}\bar{3}1m$ (D_{3d}^1) (these are part of the regular 61 space groups obtained by combining the hexagonal Bravais lattice with the various point groups). In addition, one can change the orientation of the point group elements with respect to the hexagonal basis vectors, which creates the space groups $\mathbf{P321}$ (D_3^2), $\mathbf{P31m}$ (C_{3v}^2), and $\mathbf{P}\bar{3}m1$ (D_{3d}^3).

This concludes the enumeration of the 73 symmorphic space groups.

10.3.5

The non-symmorphic space groups

The remaining 157 ($= 230 - 73$) space groups can be derived by systematically replacing one or more of the symmetry elements in the Hermann–Mauguin symbols of the point groups by screw-rotations or glide planes. Let us consider an example. We have seen in Fig. 10.6 that the combination of $\mathbf{mm2}$ (C_{2v}) with oP produces the space group $\mathbf{Pmm2}$ (C_{2v}^1). If we replace the two-fold axis by a two-fold screw axis 2_1 , then we can see from the drawing in Fig. 10.9 that a general point is transformed into seven other points, four of them at height $+$, the other four at height $\frac{1}{2} +$. Comparing this arrangement of points with the points in Fig. 10.6 we find that they are identical. In other words, if we combine the symmetry $\mathbf{mm2}_1$ with the Bravais lattice oP , then we find the symmorphic space group $\mathbf{Pmm2}$ (C_{2v}^1), with a unit cell doubled along the two-fold axis. This is not a new symmetry, so the space group $\mathbf{Pmm2}_1$ is not a new space group.

The next combination we can try is $\mathbf{mc2}_1$, i.e., we convert the mirror plane perpendicular to the $\mathbf{b}$ -axis into a c -glide plane. This combination results in the construction shown in Fig. 10.10. The presence of the c -glide plane removes half of the points that were introduced by the mirror plane in Fig. 10.9. The resulting space group is labeled $\mathbf{Pmc2}_1$ (C_{2v}^2).

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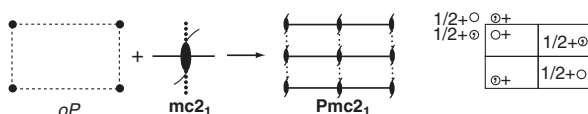


Figure 10.10 The symmetry combination $mc2_1$ gives rise to a new non-symmorphic space group $Pmc2_1$ (C_{2v}^1).

This procedure must be repeated for all possible combinations of screw axes and glide planes in all point groups and Bravais lattices. This is a tedious task and we limit ourselves to listing the resulting 157 non-symmorphic space groups in Table 10.3. Interested readers may wish to consult Buerger (1956) for a detailed and complete derivation of all 230 space groups.

Note that it is straightforward to determine which point group corresponds to a given non-symmorphic space group: simply replace all screw axes by a regular rotation of the same order, and all glide planes by a mirror plane. For instance, consider space group $I4_1/amd$ (D_{4h}^{19}). If we replace the 4_1 screw axis by a four-fold rotation, and both glides a and d by mirrors m , then we obtain point group $4/mmm$ (D_{4h}). If we use the Schoenflies notation, we can simply drop the superscript from the space group symbol.

10.3.6

*Space group generators

The section on space groups in the *International Tables for Crystallography, Volume A* is more than 600 pages long, and, as we will see in Section 10.4, contains a lot of information. The true power of group theory can be illustrated by considering what it would take to encode all space groups in a computer program. The naive way would be to key in all special positions for each space group. This would be a tremendous task, and there would be a lot of opportunities for mistakes. Group theory provides us with a much easier way of accomplishing this task. We know from the point group examples in Section 9.2.10.8 on page 216 that we can generate all the symmetry matrices of a point group starting from a list of generators. This list can be rather short for each point group, assuming that we make use of the predefined set of 14 symmetry matrices, listed in Table 9.5. Since we must also allow for translational symmetry elements in space groups, we introduce another set of symbols representing the fractions that need to be used to complete the generators elements. These symbols are shown in Table 10.4.

Since a symmetry operator is represented in matrix form by a 4×4 matrix, we must combine the generator matrices $D^{(x)}$ with the components of the translations to fully identify the space group generators. We should note that, to create the entire (infinite) space group, we must include the Bravais translation vectors as generators. We will ignore these generators in this section, since it is clear that we can go from one unit cell to another one by means of translations.

It is surprising that it is possible to compile all the information on the generators of all 230 space groups in a short ASCII file that is only 4104 bytes long! This is how it works: for each space group, we create a string of characters. The first character is either a 1 or a 0, depending on whether or not there is an inversion operator present in the group. The second

Table 10.3 The non-symmorphic space groups, with sequential number and corresponding point group.

Space group #	Point group	Symbol	Space group #	Point group	Symbol
4	$2 (C_2)$	$P2_1 (C_2^2)$	66		$Cccm (D_{2h}^{20})$
7	$m (C_s)$	$Pc (C_s^2)$	67		$Cmma (D_{2h}^{21})$
9		$Cc (C_3^4)$	68		$Ccca (D_{2h}^{22})$
11	$2/m (C_{2h})$	$P2_1/m (C_{2h}^2)$	70		$Fddd (D_{2h}^{24})$
13		$P2/c (C_{2h}^4)$	72		$Ibam (D_{2h}^{26})$
14		$P2_1/c (C_{2h}^5)$	73		$Ibca (D_{2h}^{27})$
15		$C2/c (C_{2h}^6)$	74		$Imma (D_{2h}^{28})$
17	$222 (D_2)$	$P222_1 (D_2^2)$	76	$4 (C_4)$	$P4_1 (C_4^2)$
18		$P2_12_12 (D_2^3)$	77		$P4_2 (C_4^3)$
19		$P2_12_12_1 (D_2^4)$	78		$P4_3 (C_4^4)$
20		$C222_1 (D_2^5)$	80		$I4_1 (C_4^6)$
24		$I2_12_12_1 (D_2^9)$	84	$4/m (C_{4h})$	$P4_2/m (C_{4h}^2)$
26	$mm2 (C_{2v})$	$Pmc2_1 (C_{2v}^2)$	85		$P4/n (C_{4h}^3)$
27		$Pcc2 (C_{2v}^3)$	86		$P4_2/n (C_{4h}^4)$
28		$Pma2 (C_{2v}^4)$	88		$I4_1/a (C_{4h}^6)$
29		$Pca2_1 (C_{2v}^5)$	90	$422 (D_4)$	$P42_12 (D_4^2)$
30		$Pnc2 (C_{2v}^6)$	91		$P4_122 (D_4^3)$
31		$Pmn2_1 (C_{2v}^7)$	92		$P4_12_12 (D_4^4)$
32		$Pba2 (C_{2v}^8)$	93		$P4_222 (D_4^5)$
33		$Pna2_1 (C_{2v}^9)$	94		$P4_22_12 (D_4^6)$
34		$Pnn2 (C_{2v}^{10})$	95		$P4_322 (D_4^7)$
36		$Cmc2_1 (C_{2v}^{12})$	96		$P4_32_12 (D_4^8)$
37		$Ccc2 (C_{2v}^{13})$	98		$I4_122 (D_4^{10})$
39		$Abm2 (C_{2v}^{15})$	100	$4mm (C_{4v})$	$P4bm (C_{4v}^2)$
40		$Ama2 (C_{2v}^{16})$	101		$P4_2cm (C_{4v}^3)$
41		$Aba2 (C_{2v}^{17})$	102		$P4_2nm (C_{4v}^4)$
43		$Fdd2 (C_{2v}^{19})$	103		$P4cc (C_{4v}^5)$
45		$Iba2 (C_{2v}^{21})$	104		$P4nc (C_{4v}^6)$
46		$Ima2 (C_{2v}^{22})$	105		$P4_2mc (C_{4v}^7)$
48	$mmm (D_{2h})$	$Pnnn (D_{2h}^2)$	106		$P4_2bc (C_{4v}^8)$
49		$Pccm (D_{2h}^3)$	108	$4mm (C_{4v})$	$I4cm (C_{4v}^{10})$
50		$Pban (D_{2h}^4)$	109		$I4_1md (C_{4v}^{11})$
51		$Pmma (D_{2h}^5)$	110		$I4_1cd (C_{4v}^{12})$
52		$Pnna (D_{2h}^6)$	112	$\bar{4}2m (D_{2d})$	$P\bar{4}2c (D_{2d}^2)$
53		$Pmna (D_{2h}^7)$	113		$P\bar{4}2_1m (D_{2d}^3)$
54		$Pcca (D_{2h}^8)$	114		$P\bar{4}2_1c (D_{2d}^4)$
55		$Pbam (D_{2h}^9)$	116		$P4c2 (D_{2d}^6)$
56		$Pccn (D_{2h}^{10})$	117		$P\bar{4}b2 (D_{2d}^7)$
57		$C2 (C_2^3)7$	118		$P\bar{4}n2 (D_{2d}^8)$

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Table 10.3 (cont.)

Space group #	Point group	Symbol	Space group #	Point group	Symbol
58		$Pn\bar{m}m (D_{2h}^{12})$	120		$I\bar{4}c2 (D_{2d}^{10})$
59		$Pmmm (D_{2h}^{13})$	122		$I\bar{4}2d (D_{2d}^{12})$
60		$Pbcn (D_{2h}^{14})$	124	$4/mmm (D_{4h})$	$P4/mcc (D_{4h}^2)$
61		$Pbca (D_{2h}^{15})$	125		$P4/nbm (D_{4h}^3)$
62		$Pnma (D_{2h}^{16})$	126		$P4/nnc (D_{4h}^4)$
63		$Cmcm (D_{2h}^{17})$	127		$P4/mbm (D_{4h}^5)$
64		$Cmca (D_{2h}^{18})$	128		$P4/mnc (D_{4h}^6)$
129		$P4/nmm (D_{4h}^7)$	180		$P6_222 (D_6^4)$
130		$P4/ncc (D_{4h}^8)$	181		$P6_422 (D_6^5)$
131		$P4_2/mmc (D_{4h}^9)$	182		$P6_322 (D_6^6)$
132		$P4_2/mcm (D_{4h}^{10})$	184	$6mm (C_{6v})$	$P6cc (C_{6v}^2)$
133		$P4_2/nbc (D_{4h}^{11})$	185		$P6_3cm (C_{6v}^3)$
134		$P4_2/nnm (D_{4h}^{12})$	186		$P6_3mc (C_{6v}^4)$
135		$P4_2/mbc (D_{4h}^{13})$	188	$\bar{6}m2 (D_{3h})$	$P6c2 (D_{3h}^2)$
136		$P4_2/mnm (D_{4h}^{14})$	190		$P\bar{6}2c (D_{3h}^4)$
137		$P4_2/nmc (D_{4h}^{15})$	192	$6/mmm (D_{6h})$	$P6/mcc (D_{6h}^2)$
138		$P4_2/ncm (D_{4h}^{16})$	193		$P6_3/mcm (D_{6h}^3)$
140		$I4/mcm (D_{4h}^{18})$	194		$P6_3/mmc (D_{6h}^4)$
141		$I4_1/amd (D_{4h}^{19})$	198	$23 (T)$	$P2_13 (T^4)$
142		$I4_1/acd (D_{4h}^{20})$	199		$I2_13 (T^5)$
144	$3 (C_3)$	$P3_1 (C_3^2)$	201	$m\bar{3}(T_h)$	$Pn\bar{3} (T_h^2)$
145		$P3_2 (C_3^3)$	203		$Fd\bar{3} (T_h^4)$
151	$32 (D_3)$	$P3_112 (D_3^3)$	205		$Pa\bar{3} (T_h^6)$
152		$P3_121 (D_3^4)$	206		$Ia\bar{3} (T_h^7)$
153		$P3_212 (D_3^5)$	208	$432 (O)$	$P4_232 (O^2)$
154		$P3_221 (D_3^6)$	210		$F4_132 (O^4)$
158	$3m (C_{3v})$	$P3c1 (C_{3v}^3)$	212		$P4_332 (O^6)$
159		$P31c (C_{3v}^4)$	213		$P4_132 (O^7)$
161		$R3c (C_{3v}^6)$	214		$I4_132 (O^8)$
163	$\bar{3}m (D_{3d})$	$P\bar{3}1c (D_{3d}^2)$	218	$\bar{4}3m (T_d)$	$P4_3n (T_d^4)$
165		$P\bar{3}c1 (D_{3d}^4)$	219		$F\bar{4}3c (T_d^5)$
167		$R\bar{3}c (D_{3d}^6)$	220		$I\bar{4}3d (T_d^6)$
169	$6 (C_6)$	$P6_1 (C_6^2)$	222	$m\bar{3}m (O_h)$	$Pn\bar{3}n (O_h^2)$
170		$P6_5 (C_6^3)$	223		$Pm\bar{3}n (O_h^3)$
171		$P6_2 (C_6^4)$	224		$Pn\bar{3}m (O_h^4)$
172		$P6_4 (C_6^5)$	226		$Fm\bar{3}c (O_h^6)$
173		$P6_3 (C_6^6)$	227		$Fd\bar{3}m (O_h^7)$
176	$6/m (C_{6h})$	$P6_3/m (C_{6h}^2)$	228		$Fd\bar{3}c (O_h^8)$
178	$622 (D_6)$	$P6_122 (D_6^2)$	230		$Ia\bar{3}d (O_h^{10})$
179		$P6_522 (D_6^3)$			

Table 10.4 Encoding scheme for the components of translation vectors.

$A = \frac{1}{6}$	$B = \frac{1}{4}$	$C = \frac{1}{3}$	$D = \frac{1}{2}$	$E = \frac{2}{3}$	$F = \frac{3}{4}$
$G = \frac{5}{6}$	$O = 0$	$X = -\frac{3}{8}$	$Y = -\frac{1}{4}$	$Z = -\frac{1}{8}$	

character indicates how many generators there are, in addition to the inversion operator (if present). As an example, consider the space group **Cmm2** (C_{2v}^{11}) that we encountered in the previous section. Its generator string is given by:

$$03aDDObOOOjOOO.$$

The first character is a 0, indicating that the inversion operator is not a generator. Then, the 3 indicates that there are three generator matrices. Each generator is described by four characters: the first one determines which of the 14 standard matrices is to be used (from Table 9.5). The subsequent three characters indicate the three translation components (from Table 10.4). So, for **Cmm2** (C_{2v}^{11}) the generators are *aDDO*, *bOOO* and *jOOO*. The first generator is represented by matrix $D^{(a)}$, and has translation components (*DDO*) = (1/2, 1/2, 0). This corresponds to the 4×4 matrix:

$$\mathcal{W} = \begin{pmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

We recognize this matrix as the C-centering translation. The next generator is *bOOO*, which does not have a translational component. The corresponding 4×4 matrix is

$$\mathcal{W} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

corresponding to a two-fold rotation around the *z*-axis. And, finally, the third generator is given by *jOOO*, which corresponds to

$$\mathcal{W} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

which represents a mirror plane normal to the *y*-direction.

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The last symbol in the generator string is a 0, which indicates that for this space group there is no alternative choice for the origin. If there were a second origin choice listed in the *International Tables for Crystallography, Volume A*, then this character would be a 1, and it would be followed by three more characters indicating the location of the second origin choice with respect to the first (i.e., a translation). An example of this can be found for space group **P4/nbm** (D_{4h}^3) with generator string:

$$04b000g000c000hDDO1YYO.$$

There is no inversion operator, there are four generators (meaning four four-character substrings *b000*, *g000*, *c000*, and *hDDO*), followed by a 1 and *YYO*, indicating that the second origin choice is located at $(-1/4, -1/4, 0)$ with respect to the first origin. When the complete space group is generated by a procedure similar to that described in Section 9.2.10.8, then one can change between the two origin locations by multiplying each symmetry matrix by the 4×4 matrix representing the translation $\pm(-1/4, -1/4, 0)$; the $\pm$ sign is taken to be $+$ when going from the first choice to the second and $-$ when going the opposite way.

For the highest-order space group, **Fm $\bar{3}$ m** (O_h^5), with 192 elements, we only need six generator matrices (plus the inversion operator), and the generator string is given by:

$$16aODDaDODb000c000d000e0000.$$

There are two space groups that require seven generators, **Fd $\bar{3}$ m** (O_h^7) and **Fd $\bar{3}$ c** (O_h^8). The complete listing of all 230 space group generator strings can be found in an appendix on the book's website.

Once all symmetry operators of a space group have been determined, it is a simple matter to determine the equivalent positions, starting from a general position (x, y, z) . For special positions, one would have to eliminate positions that occur twice or more in the list generated by multiplying all matrices with the general position. We leave it as an exercise for the reader to determine the complete set of symmetry matrices for the space groups **P2/c** (C_{2h}^4) and **P6 $\bar{3}$ /m** (C_{6h}^2).

10.3.7

General remarks

Thus far we have considered space groups in 3-D and plane groups in 2-D. A more general approach to symmetry theory shows that there are four different types of symmetry groups in 3-D:

- $\mathcal{G}_3^3$: the *space* groups, describing periodic 3-D structures (infinite crystals).
- $\mathcal{G}_2^3$: the *layer* groups, describing the symmetries of objects, infinitely extending in two directions but finite in the third direction (one plane is invariant).
- $\mathcal{G}_1^3$: the *rod* groups, describing the symmetries of line-type objects, infinite in one direction (one line remains invariant).
- $\mathcal{G}_0^3$: the *point* groups, describing the symmetries of finite figures in 3-D space (only one point remains invariant).

In this text, only $\mathcal{G}_0^3$ and $\mathcal{G}_3^3$ have been discussed extensively because they are of central importance for crystallography; the other groups are used in specific research fields.

In two dimensions, one distinguishes between three different types of symmetry:

- $\mathcal{G}_2^2$: the 2-D *plane* groups, describing the symmetries of 2-D planar figures (NOT planes in 3-D space!).
- $\mathcal{G}_1^2$: the 2-D *rod* groups, describing the symmetries of line figures in a plane.
- $\mathcal{G}_0^2$: the 2-D *point* groups, describing the symmetries of finite figures in a plane.

It is interesting to consider how many possible groups there are as a function of the dimension of the space in which they exist; in the following table, the number of possible groups is given for one, two, and three dimensions. In four dimensions, there are 4250 space groups $\mathcal{G}_4^4$; all of these are known and tabulated!

	3	2	1	0
3	230	80	75	32
2	—	17	7	10
1	—	—	2	2.

10.4

The *International Tables for Crystallography*

In the remainder of this book, we will use space groups to describe important crystal structures. Before doing so, it is instructive to discuss the information that is tabulated in the *International Tables for Crystallography, Volume A* (Hahn, 1996). This book presents, among many other things, a complete listing of all 230 space groups. Typical space group descriptions are shown on the following pages for the symmorphic space group **Cmm2** (C_{2v}^{11}) (Fig. 10.11), and for the non-symmorphic space group **Pmna** (D_{2h}^7) (Fig. 10.12).

The space group description consists of the following items (referring to Figs. 10.11 and 10.12):

- At the top of the page, we find the shorthand space group symbol **Cmm2**, the space group number (35), the Schoenflies symbol C_{2v}^{11} , the complete space group symbol *Cmm2*, the point group symmetry *mm2* (C_{2v}), the crystal system (*orthorhombic*), and the Patterson symmetry, which will be defined in Chapter 11. For space group **Pmna** (D_{2h}^7), the full space group symbol is given by $P_{mna}^{\frac{2}{m} \frac{2}{n} \frac{2}{a}}$.
- Next, we have a set of drawings showing the relative positions of all symmetry elements, projected along the three main directions of the orthorhombic reference frame. The drawing in the lower right corner indicates the equivalent positions with a notation similar to the one introduced earlier in this chapter. Each symmetry operator has a specific graphical symbol. We have introduced many of them in Chapter 8; for a complete listing we refer the reader to Chapter 1 of the *International Tables for Crystallography, Volume A* (Hahn, 1996).
- Note that the drawings for **Pmna** (D_{2h}^7) are significantly more complicated than those for **Cmm2** (C_{2v}^{11}). The main reason for this is the presence of multiple glide planes.

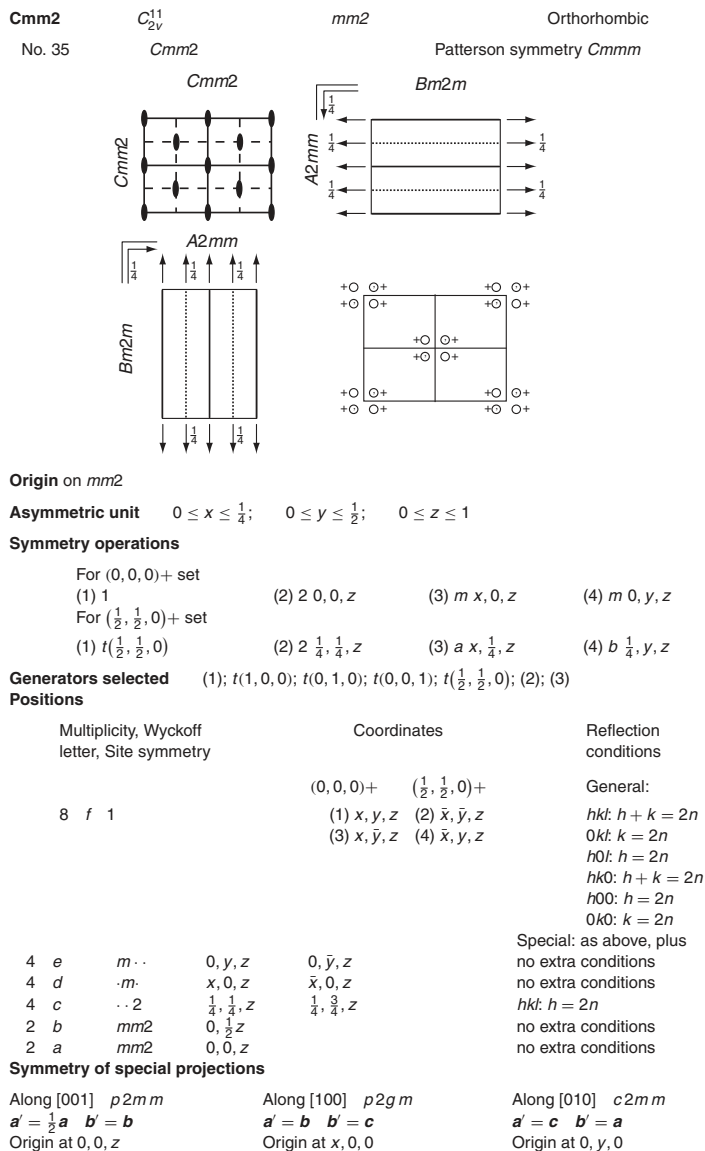
10.4 The *International Tables for Crystallography*

Figure 10.11 Example space group description for space group **Cmm2** (C_{2v}^{11}), showing all entries listed in the *International Tables for Crystallography, Volume A*, page 228 (Hahn, 1996).

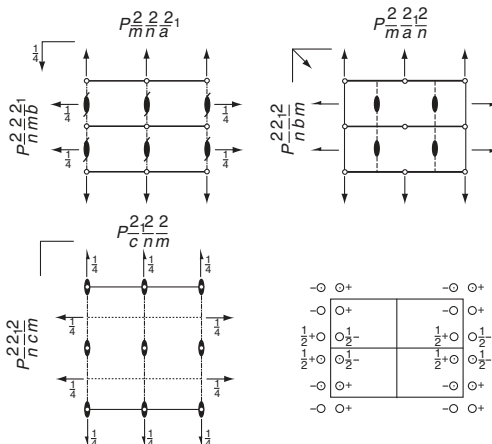
Plane groups and space groups

Pmna D_{2h}^7

mmm

Orthorhombic

No. 53

 $P2_1/m2_1/a$ Patterson symmetry $Pmmm$ Origin at center ($2/m$) at $2/mn1$ Asymmetric unit $0 \leq x \leq \frac{1}{2}; 0 \leq y \leq 1; 0 \leq z \leq \frac{1}{4}$

Symmetry operations

- (1) 1 (2) $2(0, 0, \frac{1}{2}) \frac{1}{4}, 0, z$ (3) $2\frac{1}{4}, y, \frac{1}{4}$ (4) $2x, 0, 0$
 (5) $\bar{1}0, 0, 0$ (6) $ax, y, \frac{1}{4}$ (7) $n(\frac{1}{2}, 0, \frac{1}{2})x, 0, z$ (8) $m0, y, z$

Generators selected (1); $t(1, 0, 0)$; $t(0, 1, 0)$; $t(0, 0, 1)$; (2); (3); (5)

Positions

Multiplicity,
Wyckoff
letter, Site
symmetry

Coordinates

Reflection
conditions

8 i 1

- (1) x, y, z (2) $\bar{x} + \frac{1}{2}, \bar{y}, z + \frac{1}{2}$ (3) $\bar{x} + \frac{1}{2}, y, \bar{z} + \frac{1}{2}$
 (4) $x, \bar{y}, \bar{z}$ (5) $\bar{x}, \bar{y}, \bar{z}$ (6) $x + \frac{1}{2}, y, \bar{z} + \frac{1}{2}$
 (7) $x + \frac{1}{2}, \bar{y}, z + \frac{1}{2}$ (8) $\bar{x}, y, z$

General:

 $h0l: h + l = 2n$ $hk0: h = 2n$ $h00: h = 2n$ $00l: l = 2n$ Special: as above,
plus no extra
conditions $hkt: h = 2n$ $hkt: h + l = 2n$ $hkt: h + l = 2n$ $hkt: h + l = 2n$ $hkt: h + l = 2n$ $hkt: h + l = 2n$ $hkt: h + l = 2n$

4 h m..	0, y, z	$\frac{1}{2}, \bar{y}, z + \frac{1}{2}$	$\frac{1}{2}, y, \bar{z} + \frac{1}{2}$	0, $\bar{y}, \bar{z}$
4 g .2.	$\frac{1}{4}, y, \frac{1}{4}$	$\frac{1}{4}, \bar{y}, \frac{3}{4}$	$\frac{3}{4}, \bar{y}, \frac{3}{4}$	$\frac{3}{4}, y, \frac{1}{4}$
4 f 2..	$x, \frac{1}{2}, 0$	$\bar{x} + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$	$\bar{x}, \frac{1}{2}, 0$	$x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$
4 e 2..	x, 0, 0	$\bar{x} + \frac{1}{2}, 0, \frac{1}{2}$	$\bar{x}, 0, 0$	$x + \frac{1}{2}, 0, \frac{1}{2}$
2 d 2/m..	0, $\frac{1}{2}, 0$	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$		
2 c 2/m..	$\frac{1}{2}, \frac{1}{2}, 0$	0, $\frac{1}{2}, \frac{1}{2}$		
2 b 2/m..	$\frac{1}{2}, 0, 0$	0, 0, $\frac{1}{2}$		
2 a 2/m..	0, 0, 0	$\frac{1}{2}, 0, \frac{1}{2}$		

Symmetry of special projections

Along [001] $p2mm$ Along [100] $p2gm$ Along [010] $c2mm$ $a' = \frac{1}{2}a$ $b' = b$ $a' = b$ $b' = c$ $a' = c$ $b' = a$

Origin at 0, 0, z

Origin at x, 0, 0

Origin at 0, y, 0

Figure 10.12 Example space group description for space group **Pmna** (D_{2h}^7), showing all entries listed in the *International Tables for Crystallography, Volume A*, page 268 (Hahn, 1996).

10.4 The *International Tables for Crystallography*

The arrows with a “half head” represent two-fold screw axes that lie in the plane of the drawing; the number next to the symbol indicates the height of the screw axes above the plane of the drawing.

- For space group **Cmm2** (C_{2v}^{11}), the origin of the reference frame has point symmetry **mm2** (C_{2v}), i.e., the origin is taken at the intersection of two mirror planes and a two-fold axis. For **Pmna** (D_{2h}^7), the origin is taken at a point with symmetry **2/m** (C_{2h}), where a two-fold axis (lying in the plane of the figure at upper right) intersects a mirror plane. Note that this point also coincides with an inversion operator. For many space groups, the *International Tables* list two possible origin choices; in those cases, the *Tables* show all the information in Fig. 10.11 for both origin choices.
- The *asymmetric unit* is the smallest part of space from which, by application of all symmetry operations, the whole of space is filled exactly. In other words, it is that portion of the unit cell that will completely fill all of space when it is copied by the various symmetry operators. Recall that space groups are infinite groups, by virtue of the lattice translations. The volume of the asymmetric unit is equal to the volume of the unit cell divided by the product of the order, n , of the point group corresponding to the space group, and one plus the number of centering operations. For **Cmm2** (C_{2v}^{11}), we find that $n = 4$ and there is a C-centering operation, so that the volume, V_a , of the asymmetric unit equals $V_a = V/(4 \times (1 + 1)) = V/8$. For **Pmna** (D_{2h}^7), we have $V_a = V/(8 \times 1) = V/8$.
- A complete list of all *symmetry operations*, grouped by centering vector (if any). In the case of **Cmm2** (C_{2v}^{11}), there is a C-centering operation, whereas **Pmna** (D_{2h}^7) is a primitive space group. The notation for the symmetry elements includes the location of each of the elements, and a more complete explanation can be found in Chapter 11 of the *International Tables for Crystallography, Volume A* (Hahn, 1996). There are eight symmetry operators in space group **Pmna** (D_{2h}^7); in addition to the identity, we have a two-fold screw axis, two two-fold rotation axes along the two directions normal to the screw axis, an inversion center at the origin, an a glide plane, a diagonal glide plane, and a regular mirror plane.

For each space group, a similar display can be found in the *International Tables for Crystallography, Volume A*. For tetragonal, hexagonal, and rhombohedral symmetry, the *Tables* show only one graphical representation of the symmetry operators, projected along the [001], [00.1], and [111] directions, respectively. For many of the cubic space groups, only one quadrant of the unit cell is shown, because of the large number of symmetry operators. The highest possible number of symmetry operations in a space group, not counting the Bravais lattice translation vectors, but including the centering vectors, is 192; this corresponds to the combination of the highest-order point group, **m3m** (O_h), with 48 elements, and the face-centered cubic Bravais lattice, which has three centering operations. The resulting space group is **Fm3m** (O_h^5).

On the second page of each space group description, we find additional information. The most important portion is reproduced in the second parts of Figs. 10.11 and 10.12.

- The generators are the symmetry elements from which all others can be generated by matrix multiplication. Note that the basis vectors are always among the generators, as are the centering vectors. For **Cmm2** (C_{2v}^{11}), the entire space group can be created if the three basis vectors, the centering vector, and the two-fold axis and one of the mirror

planes are provided. Every other symmetry operator of the infinite space group can be created by an appropriate matrix multiplication (using the 4×4 matrices introduced in Chapter 8).

- The *positions* table indicates the most general position and all its equivalent positions. The most general position is always indicated by x, y, z . Application of a symmetry element with number (j) from the first page of the space group description then results in the position (j) x', y', z' , where the primes indicate some combination of the coordinates x, y , and z . The first set of coordinates refers to the most general point. The number at the beginning of the top line of the table gives the multiplicity of the general position. The third symbol on the first line indicates the site symmetry of the general point, which is always equal to $1 (C_1)$. The remaining entries in the table indicate *special positions*, for which the point lies on one or more of the symmetry elements. For instance, the second entry for space group **Cmm2** (C_{2v}^{11}) indicates a point of the type $0, y, z$, which lies in the mirror plane formed by the **a** and **b** vectors. Because of this, there are only half as many equivalent points, which results in a multiplicity of $8/2 = 4$, as shown in the first column. The *site symmetry* for this position is then equal to $m \cdot \cdot$. One can progressively move the special point to the intersection of two or more symmetry elements, which further reduces the multiplicity. The highest symmetry is obtained for points along the **c**-axis, for which the site symmetry is **mm2** (C_{2v}). The third column of the table indicates the site symmetry of the general (always $1 (C_1)$) and special positions. Note that these site symmetries will always be subgroups of the point group corresponding to the space group. It is customary to refer to the special positions in the following way: one provides a letter as a label for each entry in the table, starting with the last entry (highest point symmetry). One refers to a special site by the combination of the multiplicity and the letter, as in $8f$, or $4d$. The letter is known as the *Wyckoff letter*, and the symbol $16j$ is the *Wyckoff symbol*. A crystal structure may then be described by listing the Wyckoff symbols for all the atoms in the asymmetric unit. The last column in the *Positions* table contains diffraction information, and this will be discussed in more detail in Chapter 11.
- The next section shows the *symmetry of special projections*. If the unit cell is projected along the $[001]$ direction, then the resulting 2-D figure will have the plane group symmetry **p2m m** for both space groups **Cmm2** (C_{2v}^{11}) and **Pmna** (D_{2h}^7). The 2-D unit cell will have lattice parameters **a'** and **b'** that are, in general, fractions of linear combinations of the original basis vectors. The location of the origin of the 2-D unit cell is also specified.
- There are several additional entries in the space group descriptions in the *International Tables for Crystallography, Volume A*; these are related to subgroup–supergroup relations, but we will not need them for this text. The interested reader is referred to Section 2.15 in the *Tables* (Hahn, 1996).

With the exception of the last column under the heading *Positions*, which we will discuss in the following chapters, the reader should now be able to understand the basic space group information listed in the *International Tables for Crystallography*. The *Tables* can be considered as “mandatory reading” for anyone who is serious about the study of crystallography and symmetry.

10.5

*Magnetic space groups

The last part of this chapter on plane groups and space groups deals with the concept of magnetic space groups, i.e., the groups formed when the 122 magnetic point groups, introduced in Section 9.4, are combined with the 36 magnetic Bravais lattices, described in Section 3.7. As before, only combinations of point groups and lattices that belong to the same crystal system must be considered. Nevertheless, the task of enumerating all 1651 two-color space groups is a formidable one, and was completed in the mid-1950s by Belov, Neronova, and Smirnova (Belov *et al.*, 1956; Belov and Tarkhova, 1956) and Zamorzaev (Zamorzaev, 1957); the application of two-color space groups to magnetic structures was first reported by Koptsik in 1966 (Koptsik, 1966). More recently, Litvin (Litvin, 2010) created the equivalent to the *International Tables for Crystallography, Volume A*, but for magnetic space groups.

While the magnetic space groups can be derived starting from the magnetic point groups and the magnetic Bravais lattices, an easier derivation is possible starting from the 230 crystallographic space groups discussed earlier in this chapter. For each space group, one must consider whether or not replacing half of the symmetry operators in the group by their time-reversed versions would produce a new group. When this is done systematically for all symmetry operators, including the Bravais lattice translation vectors, one finds a total of 1191 magnetic space groups. In addition, to each crystallographic space group one can add the time-reversal operator $\bar{1}'$ (i') directly, leading to 230 non-magnetic (or gray) space groups. Combining these with the 230 regular crystallographic space groups results in a total of $1191 + 230 + 230 = 1651$ magnetic (or Shubnikov) space groups.

In this section, we will, by way of illustration, consider the magnetic space groups derived from only one of the 230 regular crystallographic space groups, namely the monoclinic group $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5) (#14). This space group has a total of four symmetry operators: the identity, a two-fold screw axis parallel to the monoclinic $\mathbf{b}$ -axis through the point $(0, 0, 1/4)$, an inversion center in the origin, and a c -glide with translation vector $[001/2]$ parallel to the x - z plane through the point with coordinates $(0, 1/4, 0)$. This space group is a non-symmorphic group, derived from the symmorphic space group $\mathbf{P2}/\mathbf{m}$ (C_{2h}^1) (#10) by replacing the two-fold axis by a screw axis and the mirror plane by a glide plane. The $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5) space group symmetry elements are illustrated in Fig. 10.13(a) in 3-D and in (b) in 2-D, using the international graphical symbols for the symmetry operators. As before, the four basic symmetry elements combined with the translation vectors of the Bravais lattice generate additional inversions, screws, and glides. Note that the screw axes are located at $\mathbf{c}/4$ above the plane of the drawing.

Within the collection of magnetic space groups, each space group receives a number label that consists of three numbers separated by dots, i.e., $N_1.N_2.N_3$. The first number is the sequential number of the regular (crystallographic) space group from which the magnetic space group is derived; in this case we have $N_1 = 14$. Within the small set of magnetic space groups that can be derived from $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5), each group receives a second sequential number, in this case $N_2 = 1$. Finally, the third number is the true sequential number within the full set of 1651 magnetic space groups, in this case $N_3 = 86$. The full numerical space group label for $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5) is then 14.1.86.

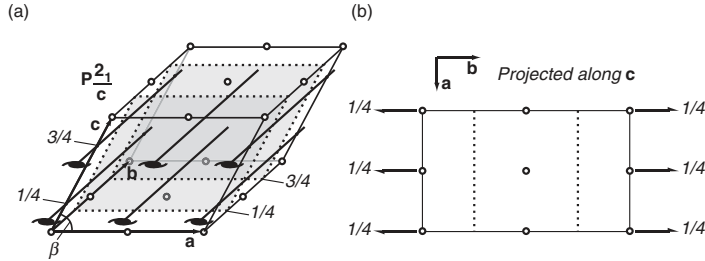


Figure 10.13 (a) 3-D schematic of the symmetry elements of the non-symmorphic space group $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5); (b) 2-D symmetry projection along the monoclinic $\mathbf{c}$ -axis.

The general point with coordinates (x, y, z) is copied onto three equivalent points by the operators 2_1 , $\bar{1}$ (i), and c of the group. In addition, since we are considering magnetic symmetry, we must determine how the direction of a magnetic moment vector placed at (x, y, z) changes when the moment vector is copied onto the other locations. This is described by adding to the position vector symbol a direction vector of the type $[u, v, w]$ (commas are added here to distinguish the moment vector components clearly). Starting from the equivalent positions in the regular crystallographic space group, and using the transformation rules for axial vectors, the equivalent positions and orientations of an axial moment vector $[u, v, w]$ at position (x, y, z) in the magnetic space group $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5) are then given by:

$$x, y, z[u, v, w]; \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[\bar{u}, v, \bar{w}]; \quad \bar{x}, \bar{y}, \bar{z}[u, v, w]; \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[\bar{u}, v, \bar{w}].$$

When we add the four moment vectors together, we obtain $[0, 4v, 0]$ as the net moment vector, which indicates that the space group $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5) can describe a ferromagnetic structure with net moment along the monoclinic $\mathbf{b}$ -axis, i.e., along the screw axis direction.

The derivation of the other magnetic space groups based on $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5) is now straightforward. First, we add the time-reversal operator $\bar{1}'$ (i') to the group, to obtain the non-magnetic (or gray) space group $\mathbf{P2}_1/\mathbf{c1}'$. Adding the time-reversal operator means that for each ordinary symmetry operator, the time-reversed version is also present; the order of the space group is then twice the order of the original one, and the eight equivalent points and moment orientations are given by:

$$\begin{aligned} x, y, z[u, v, w]; & \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[\bar{u}, v, \bar{w}]; & \quad \bar{x}, \bar{y}, \bar{z}[u, v, w]; & \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[\bar{u}, v, \bar{w}]; \\ x, y, z[\bar{u}, \bar{v}, \bar{w}]; & \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[u, \bar{v}, w]; & \quad \bar{x}, \bar{y}, \bar{z}[\bar{u}, \bar{v}, \bar{w}]; & \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[u, \bar{v}, w]. \end{aligned}$$

Adding the moment vectors together results in $[0, 0, 0]$, in agreement with the fact that there can be no net moment in groups that contain $\bar{1}'$ (i'). This gray group $\mathbf{P2}_1/\mathbf{c1}'$ receives the number label 14.2.87.

10.6 Chapter summary

The next three magnetic space groups are obtained by systematically replacing half of the symmetry operators (with the exception of 1 (E)) by their time-reversed versions. The three possibilities are $\mathbf{P2}_1'/\mathbf{c}$ (14.3.88), $\mathbf{P2}_1/\mathbf{c}'$ (14.4.89), and $\mathbf{P2}_1'/\mathbf{c}'$ (14.5.90); each of these groups has four symmetry elements, and the equivalent positions and moment directions are as follows.

For $\mathbf{P2}_1'/\mathbf{c}$:

$$x, y, z[u, v, w]; \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[u, \bar{v}, w]; \quad \bar{x}, \bar{y}, \bar{z}[\bar{u}, \bar{v}, \bar{w}]; \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[\bar{u}, v, \bar{w}];$$

with the resulting net moment $[0, 0, 0]$.

For $\mathbf{P2}_1/\mathbf{c}'$:

$$x, y, z[u, v, w]; \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[\bar{u}, v, \bar{w}]; \quad \bar{x}, \bar{y}, \bar{z}[\bar{u}, \bar{v}, \bar{w}]; \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[u, \bar{v}, w];$$

with the resulting net moment $[0, 0, 0]$.

For $\mathbf{P2}_1'/\mathbf{c}'$:

$$x, y, z[u, v, w]; \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[u, \bar{v}, w]; \quad \bar{x}, \bar{y}, \bar{z}[u, v, w]; \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[u, \bar{v}, w];$$

with the resulting net moment $[4u, 0, 4w]$, i.e., a net moment in the x - z plane.

Finally, we must consider the possibility that one of the Bravais lattice translation vectors can be replaced by its time-reversed version. From the list of magnetic Bravais lattices on page 71 we find that there are two primitive magnetic Bravais lattices, mP_a with basis vectors $\mathbf{a}'$, $\mathbf{b}$, $\mathbf{c}$ and mP_b with basis vectors $\mathbf{a}$, $\mathbf{b}'$, $\mathbf{c}$; in each case, the unit cell size is doubled along the time-reversing basis vector. Only one of these lattices gives rise to a new magnetic space group, namely $\mathbf{P2}_a\mathbf{2}_1/\mathbf{c}$ (14.6.91). The equivalent positions are given by the two sets:

$$(0, 0, 0) + \quad (1, 0, 0)' + \\ x, y, z[u, v, w] \quad \bar{x}, y + \frac{1}{2}, \bar{z} + \frac{1}{2}[\bar{u}, v, \bar{w}] \quad \bar{x}, \bar{y}, \bar{z}[u, v, w] \quad x, \bar{y} + \frac{1}{2}, z + \frac{1}{2}[\bar{u}, v, \bar{w}].$$

Since the translation $(1, 0, 0)'$ reverses the moment vectors in the second half of the unit cell, this space group does not represent ferromagnetic structures (i.e., the net moment vector vanishes). This concludes the enumeration of the six magnetic space groups that are based on the crystallographic space group $\mathbf{P2}_1/\mathbf{c}$ (C_{2h}^5). For more information, as well as the complete listing of all 1651 magnetic space groups and drawings of the symmetry elements, we refer the interested reader to Litvin (2010).

10.6

Chapter summary

In this chapter we have introduced the concepts of plane and space groups; these groups represent the unique combinations of point group symmetry and Bravais lattice translation symmetry in both 2-D and 3-D. In addition, we have shown by example how

one can derive the 1651 magnetic space groups, in which the time inversion operator is included as an additional symmetry operator. We have also explained the meaning of the different entries in the space group descriptions in the *International Tables for Crystallography, Volume A*. The 230 unique crystallographic space groups can be generated efficiently from a small set of generator matrices, combined with generator translations. The reader should now be able to understand most of the standard entries in the space group descriptions in these *Tables*; additional entries will be explained in the following chapters.

10.7

Historical notes

One of the striking developments in the field of crystallography was the fact that **William Barlow** (1845–1934), **Evgraf Stepanovich Federov** (1853–1919) and **Arthur Moritz Schoenflies** (1853–1928), men of different nationalities and technical interests, nearly simultaneously derived the existence of the 230 space groups in the 1890s.

Federov (Fig. 10.14(b)) was a Russian crystallographer. He introduced the concept of *regularity* to describe configurations of objects, and in 1885 he published *The Elements of the Study of Configurations*, an influential work on fundamental geometry (Federov, 1885). He then went on to apply the concept of regularity to the structure of the atom, which eventually led to the creation, by D. I. Mendeleev, of the periodic table of the elements. Subsequently, he applied the concept of regularity to crystal structures, which led, in 1890, to his derivation of the 230 space groups (known as *Federov groups* in the Russian literature). Around 1870, Camille Jordan, in France, had two students working on the study of continuous (Marius Sophus Lie) and discrete (Felix Klein) groups. About two decades later, Arthur Schoenflies picked up on this work, applied it to crystal structures, and published intermediate results, which were read with great interest by Federov. Federov then

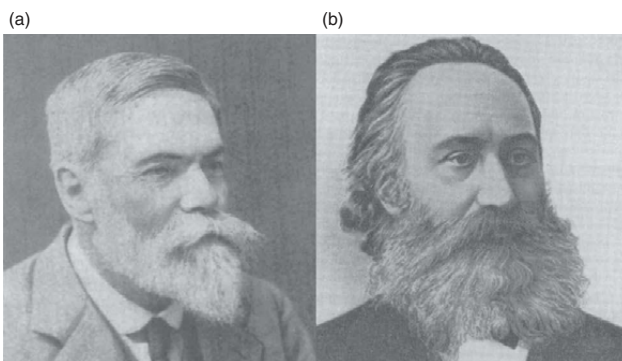


Figure 10.14 (a) William Barlow (1845–1934), and (b) Evgraf Stepanovich Federov (1853–1919) (pictures courtesy of J. Lima-de-Faria).

sent his results to Schoenflies, thus beginning a “lively correspondence” (Galiulin, 2003) between the two researchers. Federov completed his derivation and published it in 1891 (Federov, 1891), while Schoenflies published his results a few months later (Schoenflies, 1891). The reader will find an interesting account of Federov’s life and work in the article by Galiulin (2003).

Barlow (Fig. 10.14(a)) was an English amateur geologist, specializing in the field of crystallography. He derived all 230 space groups using an approach that was quite different from that used by Schoenflies and Federov (Barlow, 1894). Barlow correctly predicted the crystal structure of many compounds, including NaCl and CsCl, long before they were confirmed by means of X-ray diffraction.

By allowing each point in a crystallographic space to have a color (black or white), the 230 space groups can be extended to the 1651 Heesch–Shubnikov black–white space groups (Shubnikov and Belov, 1964). These groups can be used to describe all possible 3-D magnetic symmetries. It is also possible to extend the concept of space group to higher-dimensional spaces; it has been established that there are 4250 4-D space groups. In addition, De Wolff *et al.* (1981) published the theory of *superspace groups*, which allows for the description of commensurately and incommensurately modulated 3-D crystals in terms of 4-D groups. The study of groups as applied to crystallography is still a very active field of research, which was set in motion around 1830 by the young mathematician Evariste Galois.

10.8

Selected problems

- (i) *Plane groups*: Consider plane groups with a square lattice:
 - (a) Derive the **p4** plane group by adding the **4** (C_4) point group to a lattice point. List the equipoints, site symmetries, and multiplicities.
 - (b) Derive the **p4mm** plane group by adding the **4mm** (C_{4v}) point group to a lattice point. List the equipoints, site symmetries, and multiplicities.
- (ii) *Space groups*: The *spinel-type magnetic oxides* include *magnetite*, Fe_3O_4 , which is one of the important oxides of iron. The space group for spinel is **Fd $\bar{3}$ m** (O_h^7). This problem investigates this space group (#227). In magnetite, O^{2-} anions occupy the 32e special position with $x = 0.38$, Fe^{3+} cations occupy the 16c special position, and Fe^{2+} cations occupy the 8b special position. The magnetite lattice constant is 0.83941 nm.
 - (a) Locate the pages for space group # 227 in the *International Tables*.
 - (b) Determine the spinel Bravais lattice and list the site symmetry and equivalent positions for each ion in the Fe_3O_4 structure. How many point group operations leave each site invariant?
 - (c) Show that the site multiplicity is consistent with the Fe_3O_4 composition. How many formula units are there per unit cell?
 - (d) Using the atomic weights for Fe and O, calculate the density of Fe_3O_4 .
- (iii) *Space groups II*: Olivine is a magnetic mineral abundant in the Martian soil. Important olivines occur in the *pseudo-binary* solid solution between the isostructural

magnesium silicate (*forsterite*) and iron silicate (*fayalite*); Consider *fayalite*, which adopts space group **Pnma** (D_{2h}^{16}) (#62) with lattice constants $a = 0.4816$, $b = 1.0469$, $c = 0.6099$ nm. Its atom positions (below) are occupied by Fe, Si, and O ions. Atomic weights are (g/cm^3): $AW_{\text{Fe}} = 55.85$, $AW_{\text{O}} = 16$, and $AW_{\text{Si}} = 28.08$. Determine:

- The Bravais lattice, and Pearson symbol.
- The special positions, point group, and order at each position.
- The number of Fe, O, and Si atoms in the cell and the composition.
- The number of formula units per unit cell and the material's density.
- Which symmetry operations determine equipoints at the $8d$ position.

Atom	x	y	z
Fe(1)	0.0	0.0	0.0
Fe(2)	0.2861	0.25	0.9864
O(1)	0.092	0.25	0.7674
O(2)	0.4532	0.25	0.211
O(3)	0.1654	0.0362	0.2883
Si	0.0972	0.25	0.4305

- (iv) *Space groups III*: A secondary crystallization product of certain metallic glasses is a transition metal boride that has space group **Fm $\bar{3}$ m** (O_h^5) (#225). The cubic compound has $a = 1.059$ nm, with four inequivalent Fe atoms and one B atom in the special sites tabulated below. The atomic weights (g/cm^3) of Fe and B are: $AW_{\text{Fe}} = 55.85$ and $AW_{\text{B}} = 10.8$. Determine:

- The Bravais lattice, Pearson symbol, and point group symmetry at each special position.
- The composition, number of formula units in a unit cell, and density.
- The shortest distance between the Fe atoms on the $4a$ and $32f$ sites.

Atom	Site	x	y	z
Fe(1)	$4a$	0.00000	0.00000	0.00000
Fe(2)	$8c$	0.25000	0.25000	0.25000
Fe(3)	$32f$	0.38103	0.38104	0.38104
Fe(4)	$48h$	0.00000	0.17115	0.17115
B(1)	$24e$	0.27644	0.00000	0.00000